

Quiz Test

Sub: **Laplace Transforms and Vector Space**

Date: **21/02/2024**

Sub. Code: **CS/IS/AI/AD/CI/CY31**
Time: **4.45 – 5.00 pm**

1.	Evaluate $L\left\{t \int_0^t \frac{\sin t}{t} dt\right\}$.
2.	Find $L\{(t^2 - 6t + 9) e^{-t} U(t-3)\}$.
3.	Determine the matrix that defines a reflection in the x-axis followed by rotation through an angle $\pi/3$.
4.	Construct the orthonormal basis of the subspace spanned by $\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \end{bmatrix}\right\}$ of R^2 .
5.	Write any two properties of Eigen values and Eigen vectors.